UNIVERSITY
OF WOLLONGONG
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Faculty of Engineering and Information Sciences
School of Computing & Information Technology

CSIT321: Project

Capstone Project - Autumn 2026

User Manual

Team Members

Noor Ahmed - 7295297
Nishad Gyawali - 8124553
Chelsea Okan - 8438675

University of Wollongong

Liverpool Campus

Subject Coordinator

Dr. John Le
Dr. Khoa Nguyen

25th May, 2026

Table of Contents

Document Control

Field	Details
Document Title	SmartResearch User Manual
Project Title	SmartResearch: Intelligent Paper Summarisation and Clustering System
Subject	CSIT321: Project
Submission	Assignment 6 Final Product and Documentation
Prepared By	Chelsea Okan
Intended Users	Students, researchers, academic staff, project markers, and future maintainers
Product Type	Local Web-Based Prototype
Main Workflow	Upload → All Papers → Cluster → Export
Final Frontend Routes	/upload, /papers, /papers/:id, /cluster, /export
Backend Address	http://127.0.0.1:8000
Frontend Address	Provided by Vite after running npm run dev

Glossary

Term	Meaning
Backend	The FastAPI server that processes uploaded files, stores data, performs search and clustering, and generates export output.
Frontend	The React/Vite browser interface used to operate SmartResearch.
PDF	Portable Document Format. This is the supported upload file type for SmartResearch.
OCR	Optical Character Recognition. OCR is used as a fallback when direct PDF text extraction is insufficient.
Metadata	Paper information such as title, authors, year, DOI, venue, abstract, summary, source, confidence, and reliability where available.
Keyword Search	Search based on direct word or phrase matching.
Semantic Search	Search based on meaning-level similarity rather than exact wording.
Hybrid Search	Search that combines keyword and semantic search results.
Cluster	A generated group of related papers.
Similar Papers	Papers from the uploaded corpus that are semantically related to a selected paper.
External Recommendations	Optional related-paper suggestions retrieved through external metadata or recommendation services where available.
Export	A downloadable output file created from processed papers and clusters.
Relevance Score	A score shown in search results to indicate how strongly a result matches the user query.
Local Storage	The file-based storage used by the prototype to store uploaded PDFs, extracted text, metadata, indexes, embeddings, and generated outputs.

1. Introduction

1.1 Purpose of This Manual

This user manual explains how to use SmartResearch, an AI-assisted academic research support prototype. It provides step-by-step instructions for starting the system, uploading academic PDF papers, reviewing processed papers, searching the document collection, exploring generated clusters, exporting results, deleting stored papers, and troubleshooting common issues.

This manual is intended for users operating SmartResearch through the browser interface. It does not require programming knowledge for normal system use. Basic setup instructions are included because the final prototype runs locally through a frontend and backend server.

1.2 Product Overview

SmartResearch helps users process and organise academic papers during early literature review work. Users upload research papers as PDF files, then use the system to extract text, view paper information, search across the uploaded corpus, explore generated topic clusters, and export results for later review.

The system is organised around four main views.

View	Route	Purpose
Upload	/upload	Add academic PDF papers to the system.
All Papers	/papers	Browse uploaded papers, review metadata and summaries, search the corpus, and delete papers.
Cluster	/cluster	View automatically generated topic clusters and grouped papers.
Export	/export	Generate downloadable output from processed papers and clusters.

1.3 What SmartResearch Can Do

Function	Description
Upload Papers	Upload one or more academic PDF files through the frontend interface.
Process Papers	Extract text, attempt OCR fallback for low-text PDFs, generate metadata, and store paper information locally.
Browse Papers	View uploaded papers in the All Papers table.
View Details	Open a detail modal to inspect metadata, summary information, and available extracted text.
Search Papers	Search using keyword, semantic, or hybrid search modes.
View Clusters	Explore generated topic groups based on uploaded papers.

Generate AI summary	Generate an abstractive summary for a selected paper, summary length can be generated on demand
Recover/Replace Summary	Replace the stored extractive summary with a newly generated AI summary. The backend preserves the original extractive summary for recovery where required.
View Related Papers	Use similar-paper or recommendation features where available in the final build.
Export Results	Generate downloadable output from the processed paper collection.
Delete Papers	Remove stored papers from the local prototype.

1.4 Intended Users

User Type	**Typical Use**
Students	Organising sources for assignments, reports, capstone work, or literature reviews.
Researchers	Reviewing and grouping academic paper collections.
Academic Staff	Inspecting uploaded papers, summaries, and topic groups.
Project Markers	Testing the final prototype workflow.
Future Developers	Understanding the user-facing behaviour before extending the system.

1.5 Manual Structure

This manual is organised in the same order as the main user workflow:

System requirements.
Installation and startup.
Quick-start guide.
Uploading papers.
Viewing All Papers.
Viewing paper details.
Searching papers.
Exploring clusters.
Exporting results.
Deleting papers.
Recommended workflows.
Troubleshooting.
Known limitations.

User safety and academic good practice.
Quick reference and example walkthrough.

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2. System Requirements

2.1 Required Software

To run SmartResearch locally, the following software is required.

Requirement	Purpose
Python	Runs the FastAPI backend.
Python Virtual Environment	Isolates backend dependencies.
Node.js and npm	Runs the React/Vite frontend.
Modern Web Browser	Opens and operates the SmartResearch interface.
Git	Clones or accesses the source code repository if needed.

2.2 Recommended Environment

Item	Recommended Value
Backend Runtime	Python 3.11.x.
Backend Framework	FastAPI served by Uvicorn.
Frontend Runtime	Node.js/npm with Vite.
Browser	Chrome, Edge, Firefox, or another modern browser.
Backend URL	http://127.0.0.1:8000
Frontend URL	Provided by Vite after running npm run dev.

2.3 Supported Upload Files

File Type	Supported?	Notes
PDF	Yes	Main supported upload type.

Text-selectable academic PDF	Yes	Best supported input type.
Scanned PDF	Partially	OCR fallback may attempt extraction, but results depend on local setup and scan quality.
Word Document	No	Not supported as an upload type.
PowerPoint	No	Not supported as an upload type.
Image File	No	Not directly supported as an upload type.
Excel/CSV	No	Not supported as document upload inputs.

2.4 Prototype Assumptions

SmartResearch is a final capstone prototype. Users should understand the following assumptions before using it.

Assumption	Meaning
Local Prototype	The system runs locally using separate backend and frontend servers.
Single-user Workflow	The prototype does not include user accounts, authentication, or role-based access.
Local Storage	Uploaded papers and processed outputs are stored locally.
Academic Paper Input	The system is designed for academic PDF papers, not arbitrary document types.
Processing Time Varies	File size, OCR use, semantic indexing, and local machine performance affect processing time.
Generated Output Requires Review	Summaries, metadata, search results, recommendations, and clusters should be checked by the user before academic use.

3. Installation and Startup

3.1 Startup Overview

SmartResearch requires two local services.

Service	Purpose
Backend	Runs the FastAPI API and handles processing.
Frontend	Runs the React/Vite user interface in the browser.

The backend should be started before using the frontend. Upload, search, clustering, metadata, recommendation, deletion, and export features rely on backend API availability.

3.2 Start the Backend

Open a terminal in the project directory and run:

```
cd backend
python3 -m venv .venv
source .venv/bin/activate     # Windows: .venv\Scripts\activate
pip install -r requirements.txt
uvicorn app:app --reload
```

Expected result:

Expected Output	Meaning
Uvicorn server starts successfully.	Backend is running.
Backend available at `http://127.0.0.1:8000`.	Frontend can call backend endpoints.
Swagger UI available at `http://127.0.0.1:8000/docs`.	Backend API documentation can be viewed.

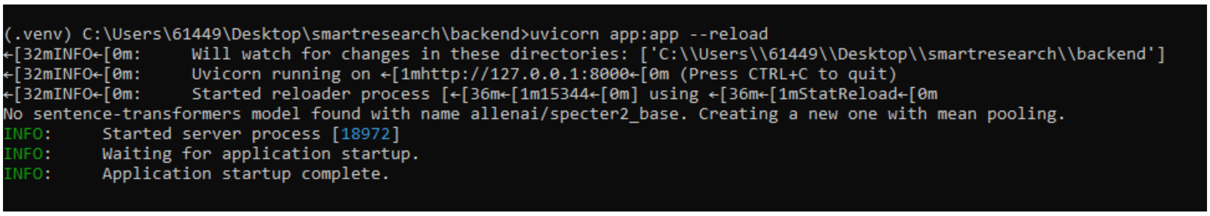

Figure 1: Backend Terminal Running Successfully

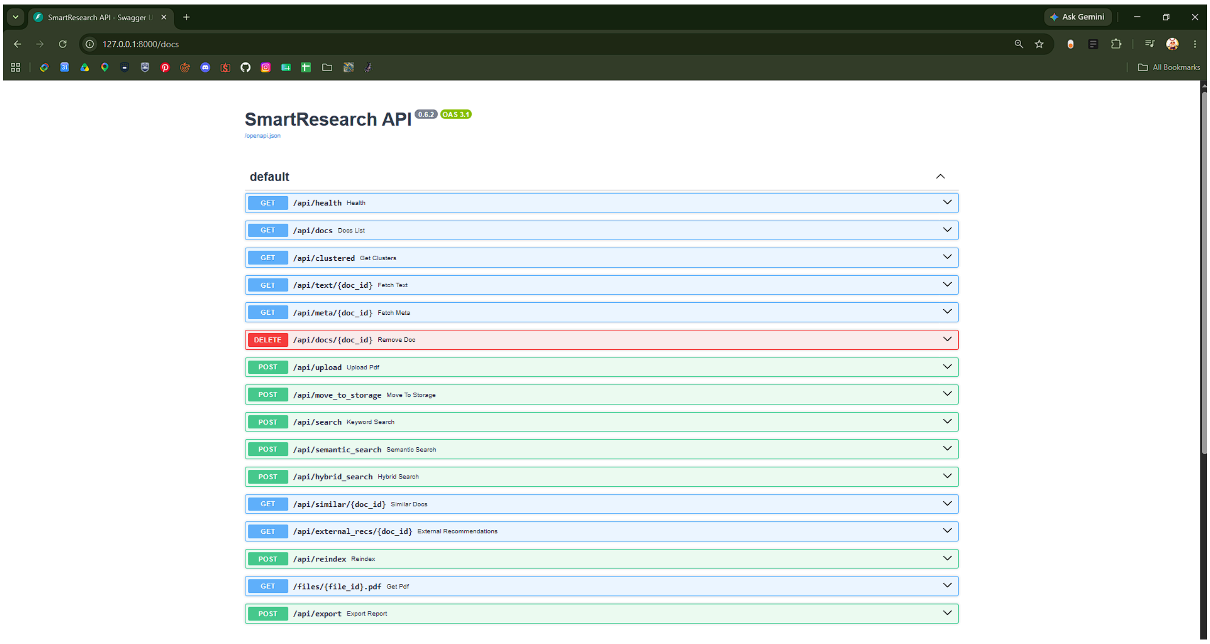

Figure 2: FastAPI Swagger UI

3.3 Check Backend Health

After starting the backend, open:

http://127.0.0.1:8000/api/health

Expected result:

Result	**Meaning**
Health response appears.	Backend is available.
Error or page not found.	Backend may not be running correctly, or the final backend uses a different health/status endpoint.

Figure 3: Backend Health Response

3.4 Start the Frontend

Open a second terminal in the project directory and run:

```
cd frontend
npm install
npm run dev
```

Expected result:

Expected Output	**Meaning**
Vite development server starts.	Frontend is running.
Local URL appears in terminal.	Browser can open the SmartResearch interface.
Upload page loads.	System is ready for user operation.

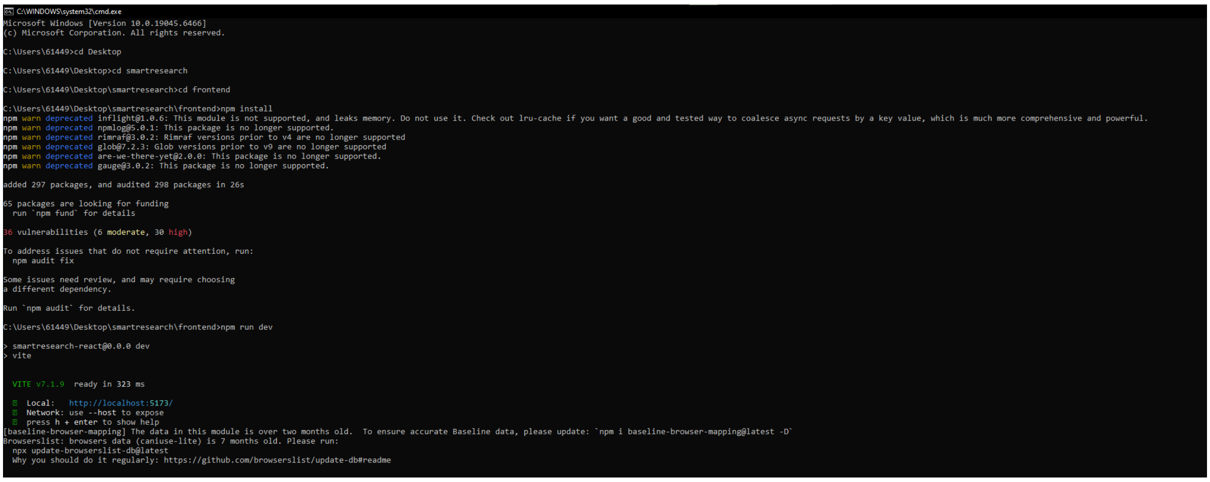

Figure 4: Frontend Terminal Running Successfully

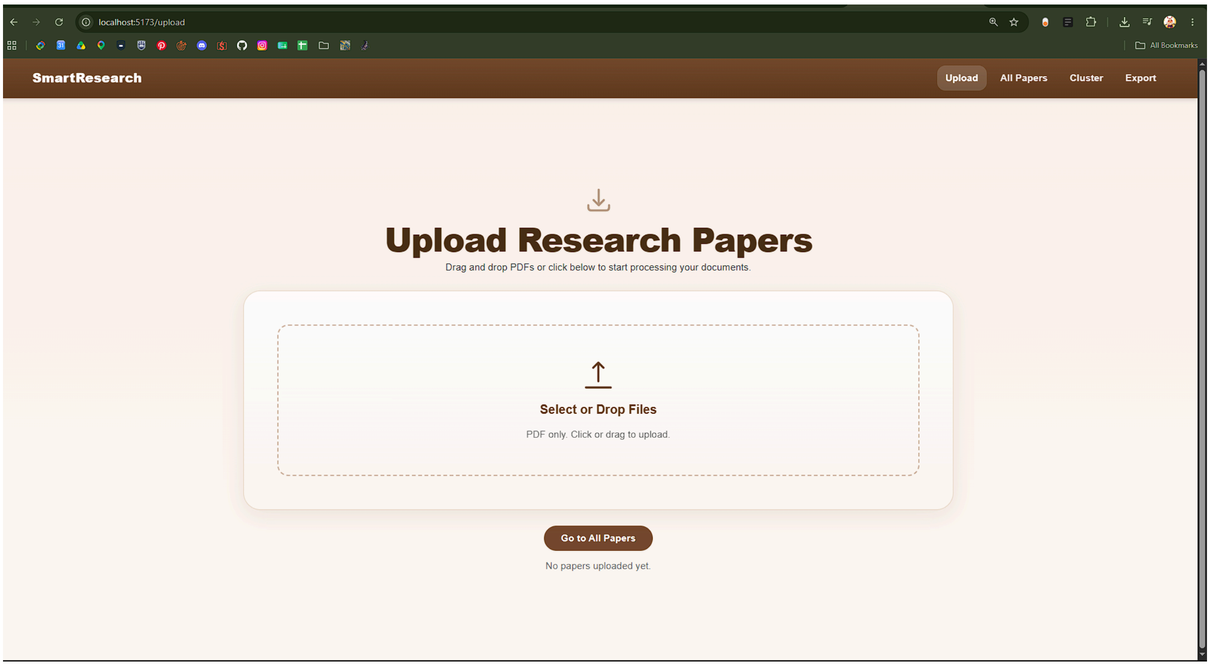

Figure 5: SmartResearch Upload Page Loaded

3.5 Startup Checklist

Before using SmartResearch, confirm:

Check	Complete
Backend terminal is running.	☐
Frontend terminal is running.	☐
Browser opens the SmartResearch frontend.	☐
Upload page is visible.	☐
Backend health endpoint or confirmed API endpoint responds.	☐
Swagger UI is accessible where available.	☐

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4. Quick-Start Guide

4.1 Fastest Way to Use SmartResearch

Use this quick-start guide when the backend and frontend are already running.

Step	Action	Expected Result
1	Open the SmartResearch frontend.	Upload page appears.
2	Upload one or more academic PDF files.	Files appear in the upload list.
3	Wait for upload and processing to complete.	Files show completed status.
4	Click Go to All Papers or navigate to All Papers.	All Papers page opens.
5	Review uploaded papers in the table.	Paper records, summaries, and metadata appear where available.
6	Search using Hybrid, Semantic, or Keyword mode.	Matching papers appear with relevance scores where available.
7	Navigate to Cluster.	Cluster page opens.
8	Open a cluster to view grouped papers.	Cluster detail modal appears.
9	Navigate to Export.	Export page opens.
10	Select export options and click Export.	Export file downloads if backend export succeeds.

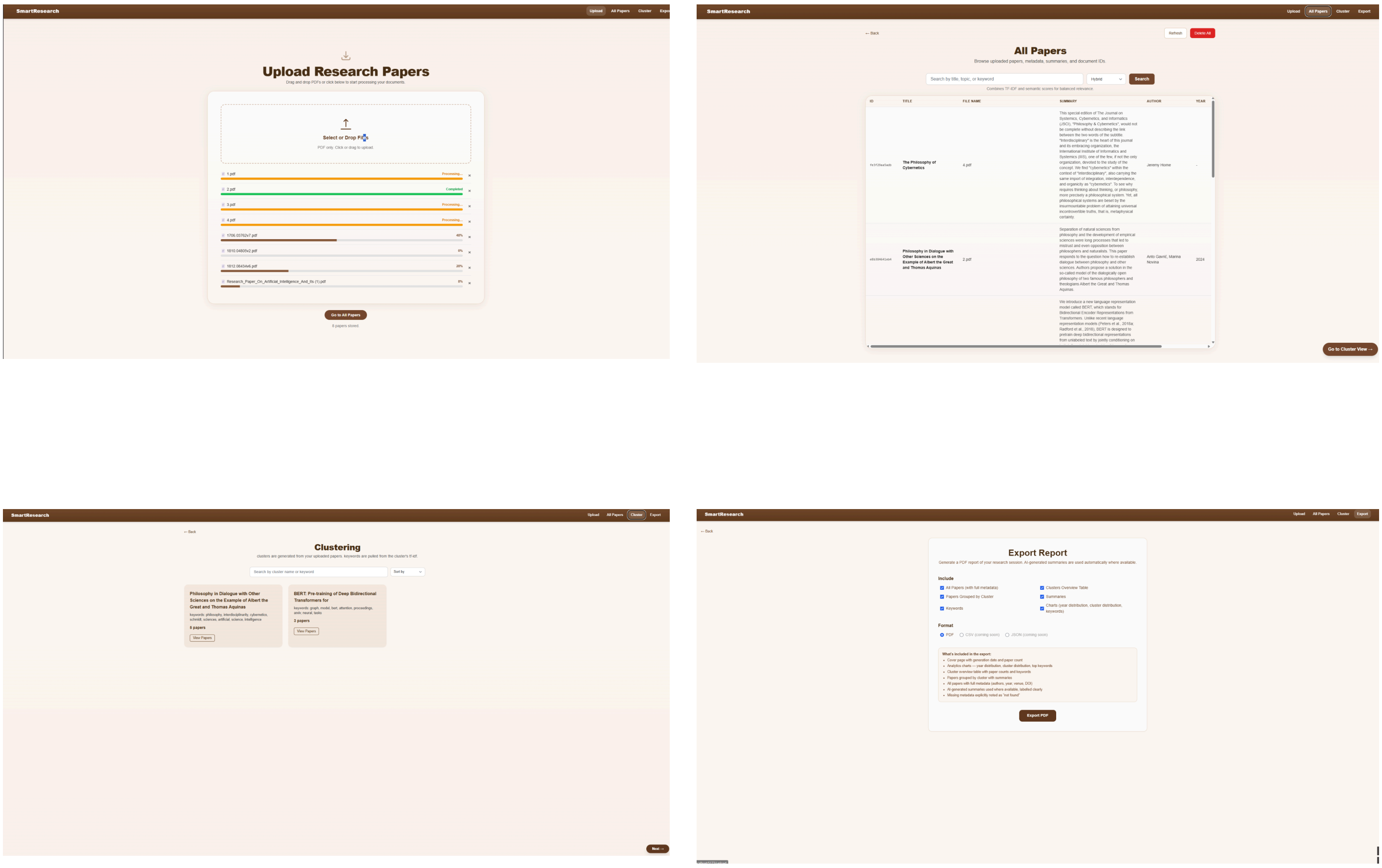

Figure 6: SmartResearch Core User Workflow

4.2 Recommended First-Time Test

For first-time use, upload two or three text-selectable academic PDFs first. This makes it easier to confirm that upload, browsing, searching, clustering, and export work before attempting a larger document batch.

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5. Uploading Papers

5.1 Purpose of the Upload Page

The Upload page is the entry point for SmartResearch. It allows users to add academic PDF files to the system. After upload, the backend processes each file and makes it available in the All Papers, Search, Cluster, Recommendation, and Export workflows where supported.

The Upload page includes the following interface elements.

Interface Element	Purpose
Upload Area	Select or drag PDF files into the system.
Upload List	Shows selected or uploaded files.
Progress Bar	Displays upload and processing progress.
Status Label	Shows uploading, processing, completed, or failed states.
PDF Preview	Opens completed uploads in a preview modal where available.
Remove Button	Removes a file from the current upload list.
Go to All Papers Button	Moves the user to the paper review page.

5.2 Upload a Single PDF

Step	Action	Expected Result
1	Open the Upload page.	Upload interface is visible.
2	Click the upload area.	File picker opens.
3	Select a PDF file.	File appears in the upload list.
4	Wait while the system uploads and processes the file.	Progress bar updates.
5	Confirm the file status changes to completed.	File is ready for review.
6	Click Go to All Papers.	Uploaded paper appears in the All Papers view.

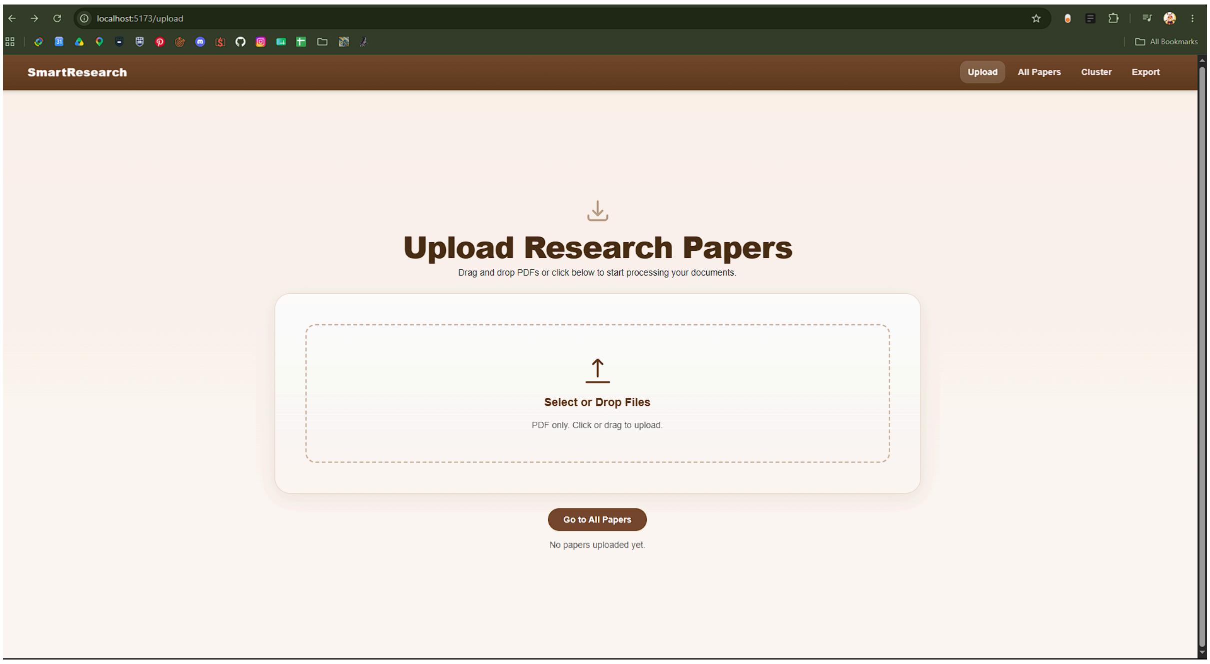

Figure 7: Upload Page Before Upload

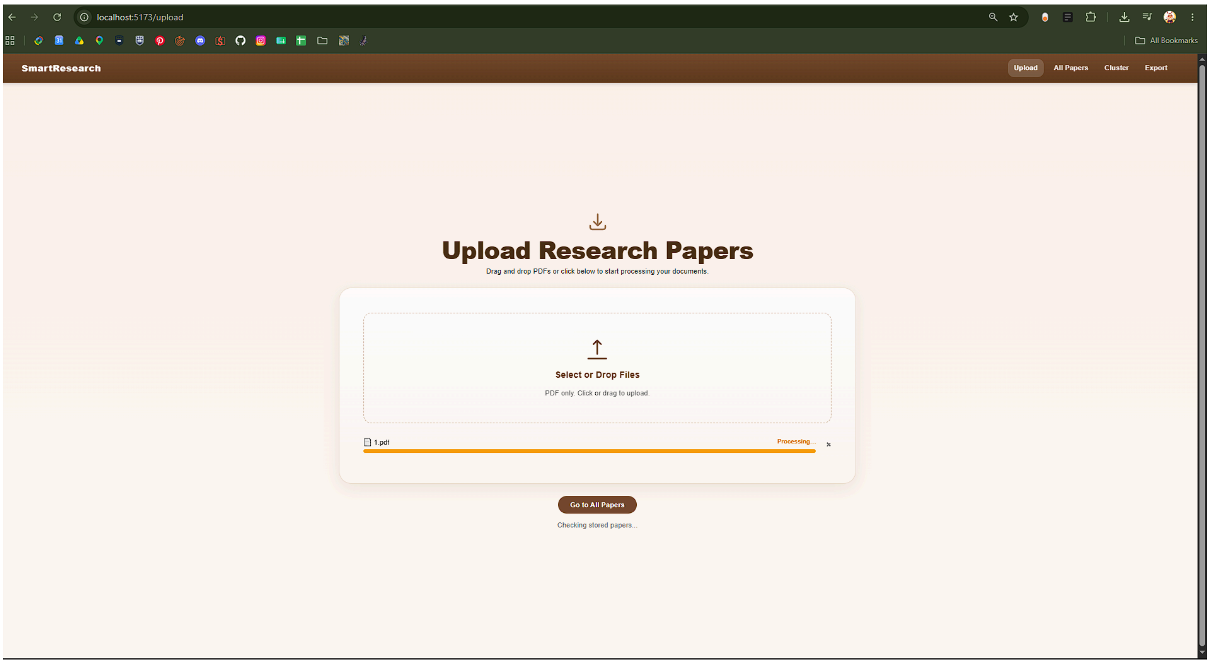

Figure 8: Single File Upload in Progress

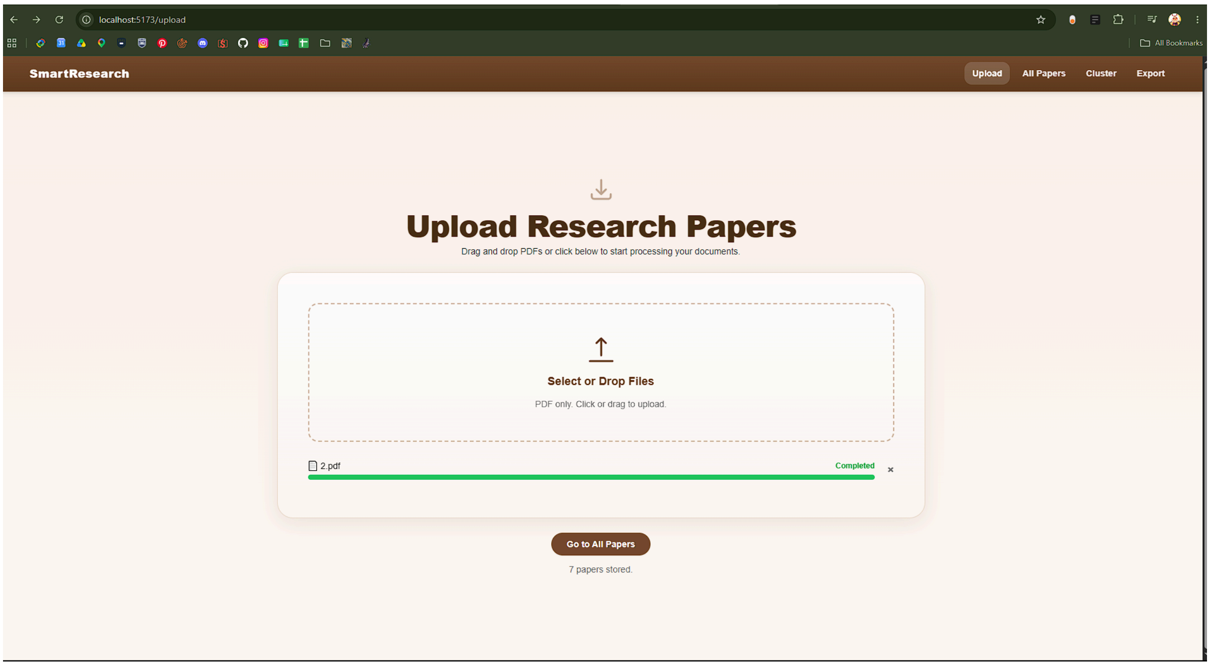

Figure 9: Single File Upload Completed

5.3 Upload Multiple PDFs

Step	Action	Expected Result
1	Click the upload area or drag multiple PDFs into it.	All selected files appear in the upload list.
2	Wait for each file to upload and process.	Each file shows a status.
3	Confirm completed files.	Successfully processed files show completed.
4	Open All Papers.	Uploaded papers appear in the table.

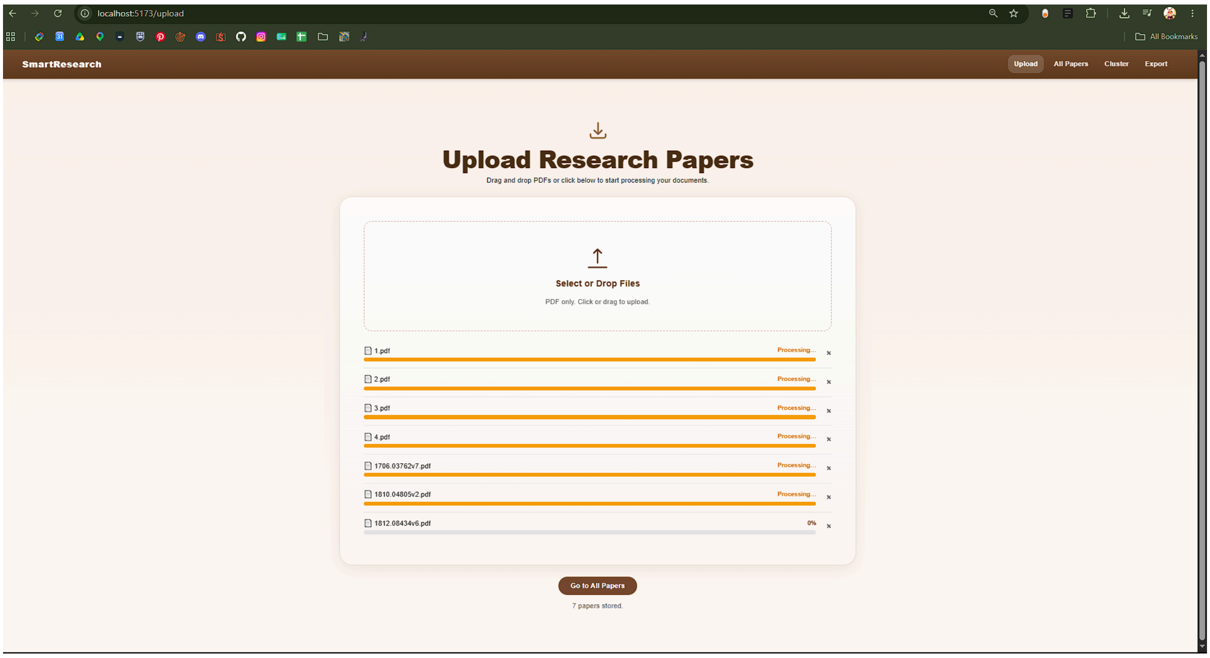

Figure 10: Multiple File Upload

5.4 Drag-and-Drop Upload

Users can drag PDF files directly into the upload box.

Step	Action	Expected Result
1	Open the folder containing the PDFs.	PDF files are visible on the computer.
2	Drag the files into the upload area.	Upload list updates.
3	Release the files.	Upload begins.
4	Wait for processing to complete.	Completed files are available for review.

5.5 Upload Status Meanings

Status	Meaning
Uploading	The file is being sent from frontend to backend.
Processing	Upload finished, and backend processing is occurring.
Completed	Upload and processing completed successfully.
Failed	Upload or processing failed.

5.6 Preview Uploaded PDFs

Completed uploads can be previewed from the upload list where preview remains available in the final build.

Step	Action	Expected Result
1	Click the file name of a completed upload.	PDF preview modal opens.
2	Review the document.	PDF is displayed in the modal.
3	Click the close button.	Modal closes and Upload page returns.

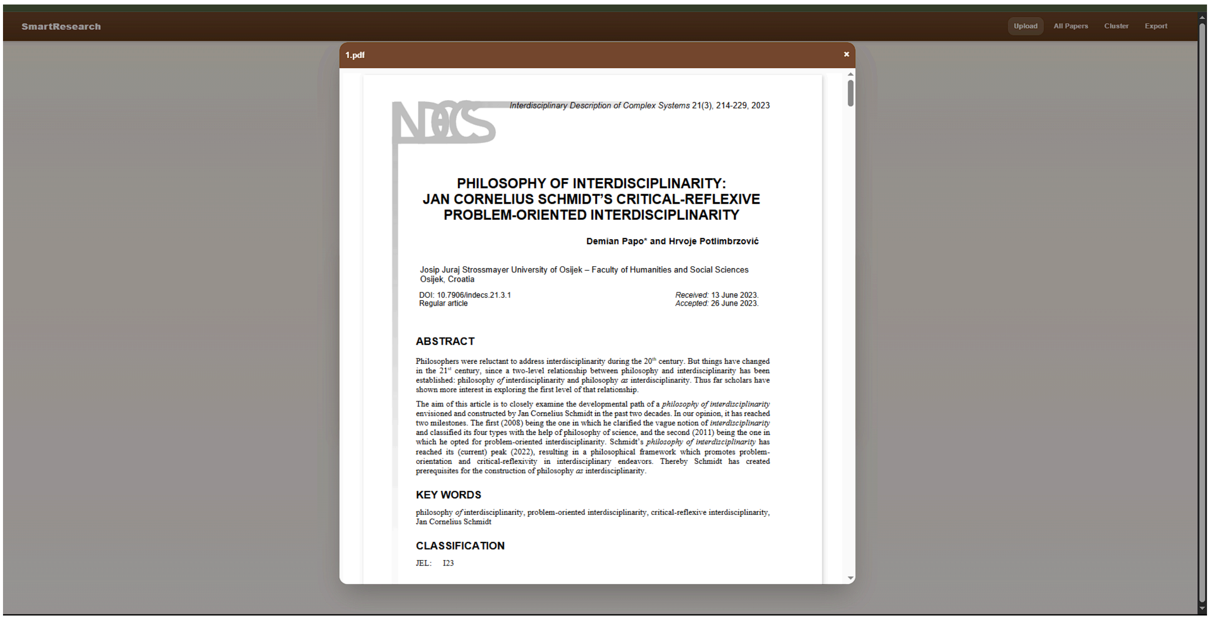

Figure 11: PDF Preview Modal

5.7 Remove a File from the Upload List

The Upload page allows users to remove a file from the upload list using the remove button or × button where available.

Step	Action	Expected Result
1	Find the uploaded file in the upload list.	File row is visible.
2	Click the remove or × button beside the file.	File is removed from the frontend upload list.

Note: Removing a file from the upload list is not the same as deleting a stored paper from the backend. To delete stored documents, use the All Papers page.

5.8 Upload Notes

SmartResearch works best with academic PDFs that contain selectable text. Scanned or image-heavy PDFs may rely on OCR fallback, which may take longer and may not extract text perfectly.

6. Viewing All Papers

6.1 Purpose of the All Papers Page

The All Papers page is the main review area for uploaded papers. It displays stored documents in a table and allows users to browse, search, inspect, refresh, and delete papers.

The All Papers page includes the following interface elements.

Interface Element	Purpose
Back Button	Returns to the previous page where available.
Refresh Button	Reloads the stored paper list.
Search Bar	Allows users to search by title, topic, or keyword.
Search Mode Selector	Selects Hybrid, Semantic, or Keyword search.
Search Button	Runs the selected search.
Paper Table	Displays uploaded papers and associated information.
Paper Title Button	Opens paper detail modal.
Delete Button	Deletes a stored paper.
Go to Cluster View Button	Moves to Cluster page where available.

6.2 Open All Papers

Step	Action	Expected Result
1	Click Go to All Papers from Upload page, or navigate to `/papers`.	All Papers page opens.
2	Wait for papers to load.	Table appears.
3	Review the displayed rows.	Uploaded papers are visible.

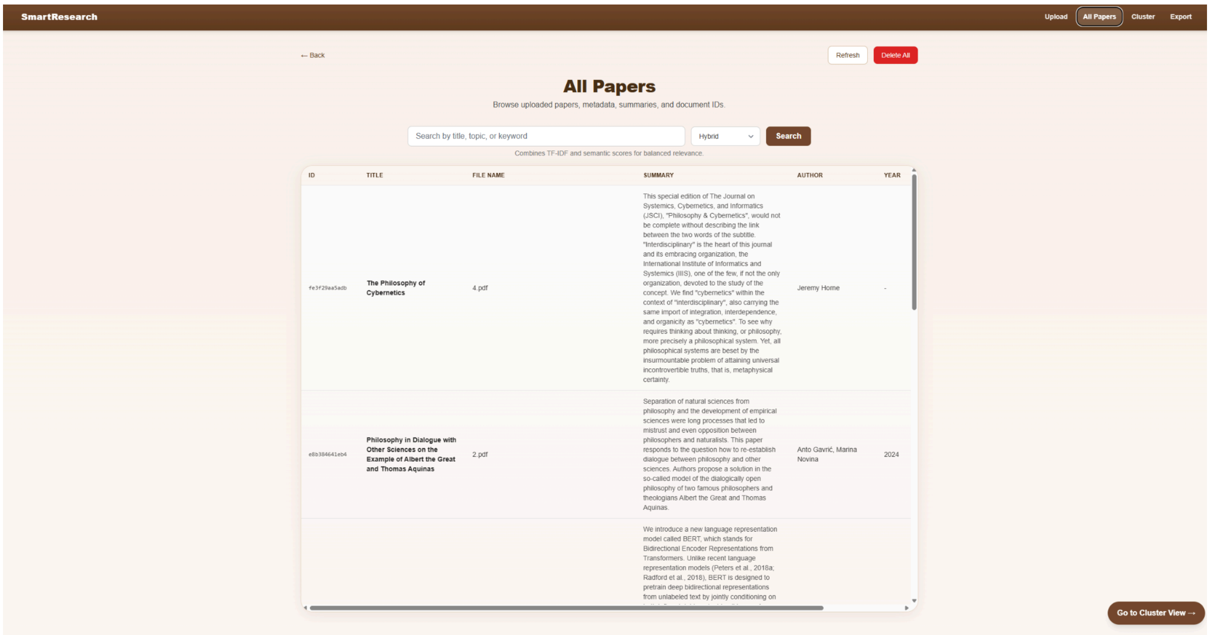

Figure 12: All Papers Page

6.3 All Papers Table Columns

Column	Description
ID	Internal document ID used by the backend where displayed.
Title	Paper title from metadata, or fallback title.
File Name	Original uploaded filename.
Summary	Generated summary or fallback summary text.
Author	Author names where available.
Year	Publication year where available.
Relevance	Search score when viewing search results.
Delete	Button used to remove the paper from storage.

6.4 Refresh the Paper List

Step	Action	Expected Result
1	Click Refresh.	System reloads papers from backend.
2	Wait for update.	Latest stored papers appear.

Use Refresh after uploading, deleting, or changing backend data.

6.5 Empty Table Message

If no papers are stored, the All Papers page may show a no-records or empty-state message.

Cause	Action
No papers have been uploaded.	Return to Upload page and upload PDFs.
Backend storage is empty.	Upload new test files.
Backend is not running properly.	Restart backend and refresh the page.

7. Viewing Paper Details

7.1 Purpose of the Detail Modal

The detail modal allows users to inspect an individual paper without leaving the All Papers page. It displays key metadata and summary information for the selected paper.

7.2 Open Paper Details

Step	Action	Expected Result
1	Open All Papers.	Paper table appears.
2	Click a paper title.	Detail modal opens.
3	Review the metadata and summary.	Paper details are visible.
4	Click × or Close.	User returns to All Papers.

Figure 13: Paper Detail Modal

7.3 Detail Modal Fields

Field	Description
Title	Paper title where available.
ID	Internal document ID.
File	Original uploaded filename.
Authors	Author names where available.
Year	Publication year where available.
Venue	Publication venue where available.
DOI	Digital Object Identifier where available .Displayed as a clickable link where supported.
Summary	Generated summary or fallback summary.
Extracted Text	Extracted paper text where available in the final interface.
Metadata Source	Indicates whether metadata came from the PDF, external lookup, or final selected metadata where displayed.
Confidence / Reliability	Indicates metadata confidence or reliability where exposed in the final interface.
Relevance	Search relevance score where applicable.

7.4 Missing Metadata

Some papers may not contain complete metadata. If a title, author, year, venue, DOI, abstract, or summary is missing, SmartResearch may show a fallback value such as the filename, a dash, or “No summary available.”

This does not always mean upload failed. It may mean the original PDF did not contain clean extractable metadata, the paper layout was difficult to parse, or external metadata enrichment could not find a reliable match. Users should check important bibliographic details against the original paper before relying on them for academic work.

7.5 Generating an Abstractive Summary

The Full Summary page allows users to generate an abstractive summary for a selected paper on demand. This feature is optional and is not part of the main workflow. It uses an AI model to produce a summary from the document and allows the user to replace the existing summary with the newly generated version.

Action	**Result**
Open All Papers	The paper table appears.
Click a paper title	The paper detail modal appears.
Click Generate Summary	The Full Summary page appears.
Select a summary length	The Short, Medium, or Long option is selected.
Click Generate Summary	AI summary generation begins.
GPU not detected	A warning appears explaining that summary generation will run on CPU and may take longer.
Processing completes	The generated summary appears on the page.
Click Generate Again	A new summary is generated using the selected summary length.
Click Replace Existing Summary	The generated summary replaces the existing summary in the All Papers view.
Click Back	The user returns to the previous page.

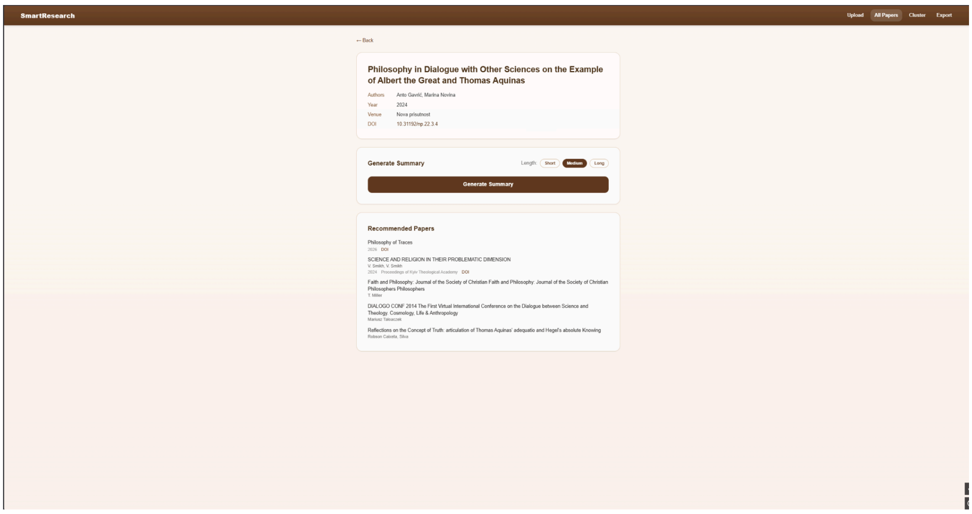

Figure 13: Abstractive Summary Generation Page

8. Searching Papers

8.1 Purpose of Search

Search helps users find relevant papers inside the uploaded collection. SmartResearch supports keyword, semantic, and hybrid search so users can retrieve papers by exact wording, meaning-level similarity, or a combined ranking approach.

Search Mode	Purpose
Keyword	Finds papers using direct word or phrase matching.
Semantic	Finds papers using meaning-level similarity rather than exact wording.
Hybrid	Combines keyword and semantic search for balanced retrieval.

8.2 Search Mode Details

Mode	Best Used When	Example Use
Keyword	The user knows an exact term, title phrase, method, or keyword.	Searching “BERT” or “graph neural network”.
Semantic	The user knows the concept but not the exact wording.	Searching “papers about attention mechanisms”.
Hybrid	The user wants a balanced general-purpose search.	Searching a topic phrase where exact and meaning-based matches both matter.

8.3 Run a Search

Step	Action	Expected Result
1	Open All Papers.	Paper table appears.
2	Type a query into the search bar.	Query appears in the input field.
3	Select Hybrid, Semantic, or Keyword from the dropdown.	Selected search mode is active.
4	Click Search.	Search results appear.
5	Review relevance scores.	Higher scores indicate stronger matches where scores are displayed.

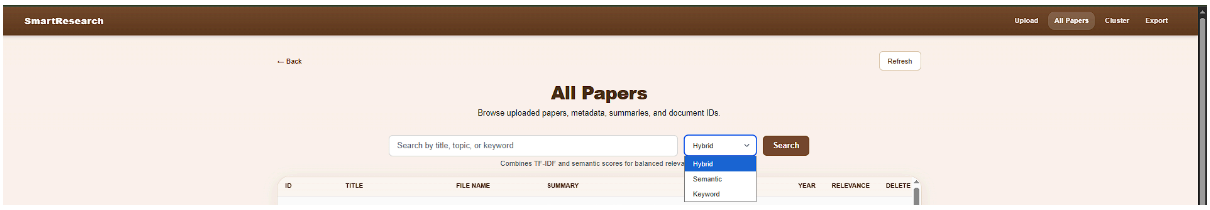

Figure 14: Search Bar and Mode Selector

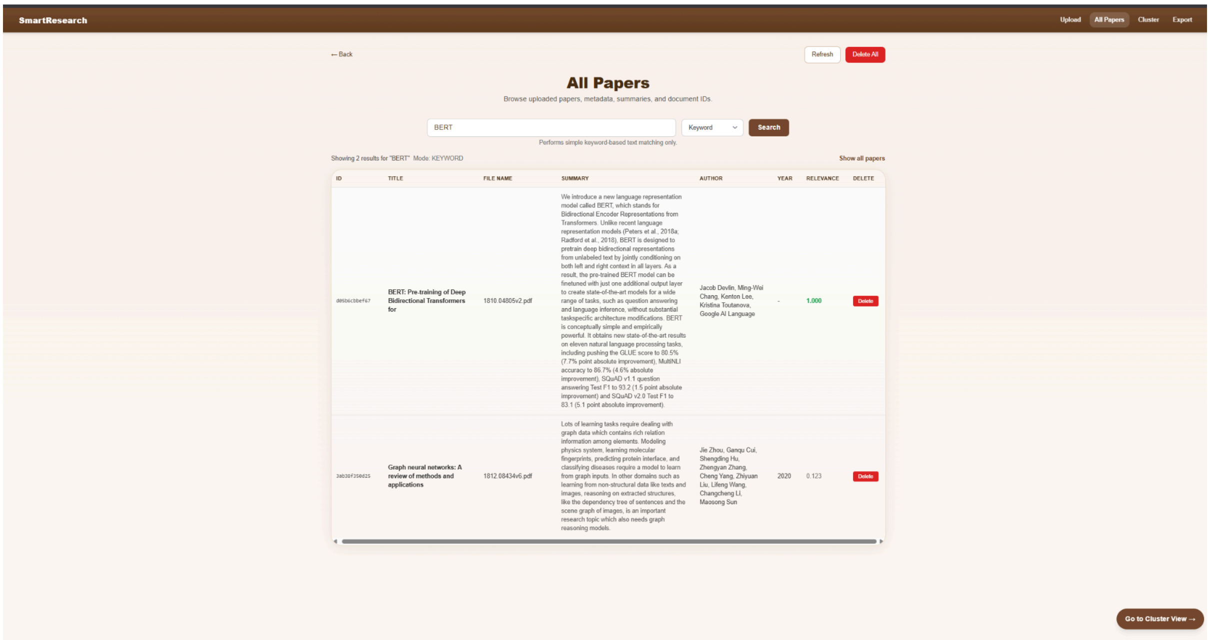

Figure 15: Search Results with Relevance Scores

8.4 Search Mode Descriptions in the Interface

The All Papers page may display short explanations for each selected search mode.

Mode	Interface Description

Semantic	Finds meaning-level matches using SPECTER2 embeddings.
Hybrid	Combines TF-IDF and semantic scores for balanced relevance.
Keyword	Performs keyword-based text matching.

8.5 Interpreting Relevance Scores

Score Type	**Meaning**
Higher score	Stronger match to the search query.
Lower score	Weaker or less direct match.
Dash or blank	Relevance is not applicable, usually when browsing All Papers instead of viewing search results.

Relevance scores are a guide only. Users should still inspect paper details and original source documents before relying on search output for academic work. Semantic and hybrid results may identify conceptually related papers even when the exact search terms do not appear in the title or summary.

8.6 Return to All Papers After Search

After running a search, the interface may show a Show All Papers option.

Step	**Action**	**Expected Result**
1	Click Show All Papers.	Search query clears.
2	Wait for reload.	Full uploaded paper table returns.

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9. Exploring Clusters

9.1 Purpose of the Cluster Page

The Cluster page shows automatically generated topic groups based on uploaded papers. It helps users move from reviewing individual papers to identifying broader themes across the paper collection.

The Cluster page includes the following interface elements where available in the final build.

Interface Element	**Purpose**
Cluster Search Bar	Filters clusters by name, description, or keyword where supported.
Sort Dropdown	Sorts clusters by name or paper count where supported.
Cluster Cards	Displays generated cluster information.
Keywords	Shows generated terms for each cluster.
Paper Count	Shows how many papers are in each cluster.
View Papers Button	Opens cluster details.
Next Button	Moves to Export page where available.

9.2 Open the Cluster Page

Step	**Action**	**Expected Result**
1	Upload and process papers first.	Papers are available in the system.
2	Click Go to Cluster View from All Papers, or navigate to `/cluster`.	Cluster page opens.
3	Wait for clusters to load.	Cluster cards appear.

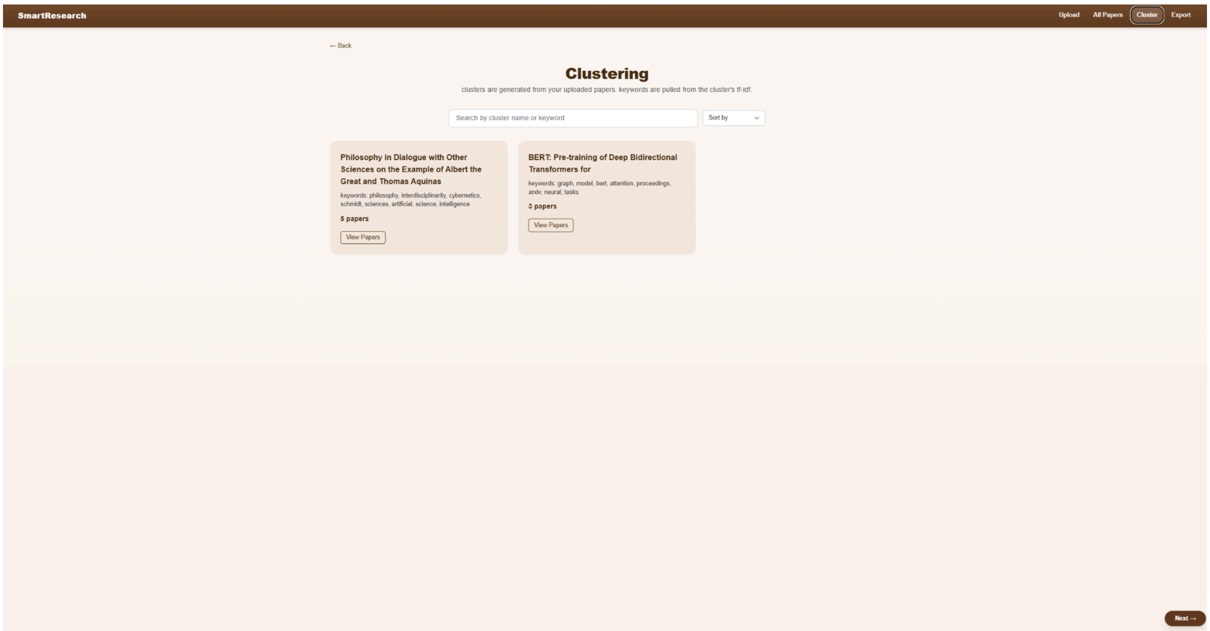

Figure 16: Cluster Page

9.3 Cluster Card Information

Cluster Field	Description
Title	Generated cluster title or fallback title.
Keywords	Important terms associated with the cluster.
Paper Count	Number of papers grouped in that cluster.
Description	Generated cluster description or keyword text.
View Papers	Opens the cluster detail modal.

9.4 Search Clusters

Step	Action	Expected Result
1	Type into the cluster search bar where available.	Cluster cards filter by matching name, description, or keyword.
2	Review visible cards.	Matching clusters remain visible.
3	Clear the search text.	Full cluster list returns.

9.5 Sort Clusters

Sort Option	Behaviour
Default	Default cluster ordering.
Name	Sorts clusters alphabetically by title where supported.
Paper Count	Sorts clusters by number of papers where supported.

9.6 Open Cluster Details

Step	Action	Expected Result
1	Click a cluster card or View Papers.	Cluster modal opens.
2	Review top keywords and paper count.	Cluster information appears.
3	Review grouped papers.	Papers assigned to the cluster appear.
4	Close the modal when finished.	User returns to Cluster page.

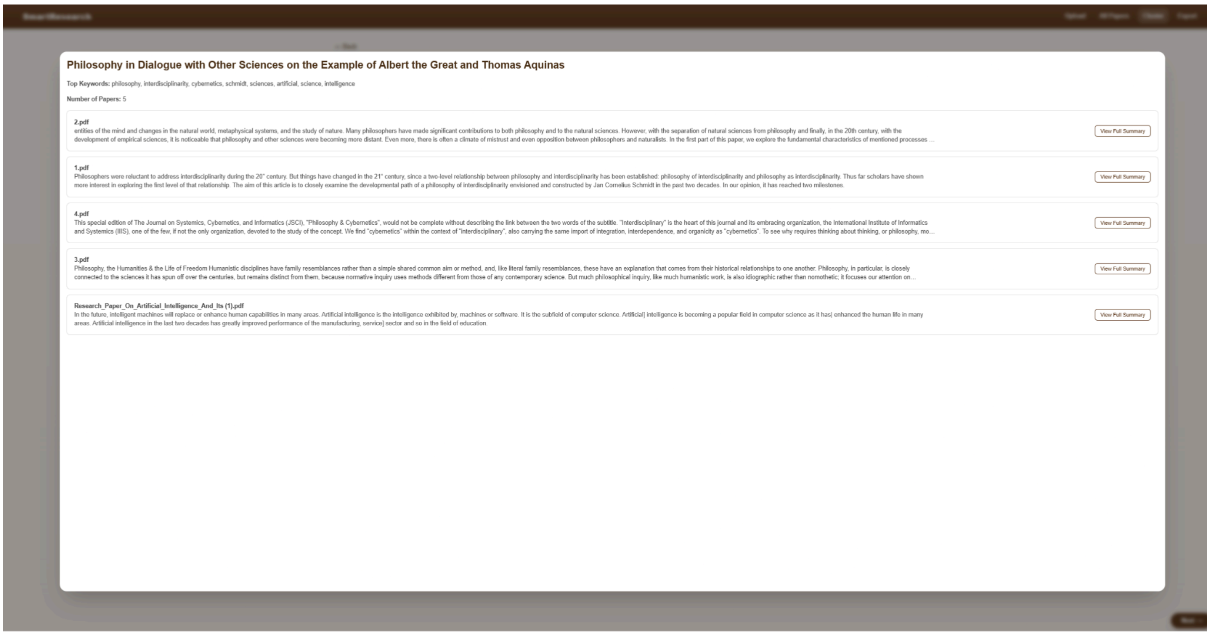

Figure 17: Cluster Detail Modal

9.7 Disabled Controls in Cluster Modal

The cluster modal may include disabled search or sort controls inside the modal.

Control	Current Behaviour
Search Papers input	Displayed but disabled where applicable.
Sort by Date dropdown	Displayed but disabled where applicable.

These controls indicate possible future functionality but should not be treated as working features unless they are implemented before final submission.

9.8 Interpreting Clusters

Clusters are generated automatically from the uploaded paper collection. Users should treat cluster labels and keywords as research aids, not final academic classifications.

Cluster Output	User Interpretation
Cluster title	Generated topic label or fallback title.
Keywords	Important generated terms, not guaranteed perfect.
Paper count	Number of papers assigned to the cluster.
Grouped papers	Papers that the backend grouped together based on available document data.
Other cluster	Papers that did not fit closely enough into any generated cluster group

Users should inspect grouped papers manually before relying on a cluster for formal research conclusions. Cluster quality depends on the number of uploaded papers, extracted text quality, metadata quality, and how similar the uploaded papers are to one another.

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10. Exporting Results

10.1 Purpose of Export

The Export page allows users to generate downloadable output from the processed paper collection. Export supports the final stage of the workflow by allowing users to reuse processed paper and cluster information outside the SmartResearch interface.

10.2 Open the Export Page

Step	Action	Expected Result
1	Click Next from Cluster page, or navigate to `/export`.	Export page opens.
2	Review export options.	Content and format options are visible.
3	Select desired options.	Checkboxes and format selection update.
4	Click Export.	File download begins if backend export succeeds.

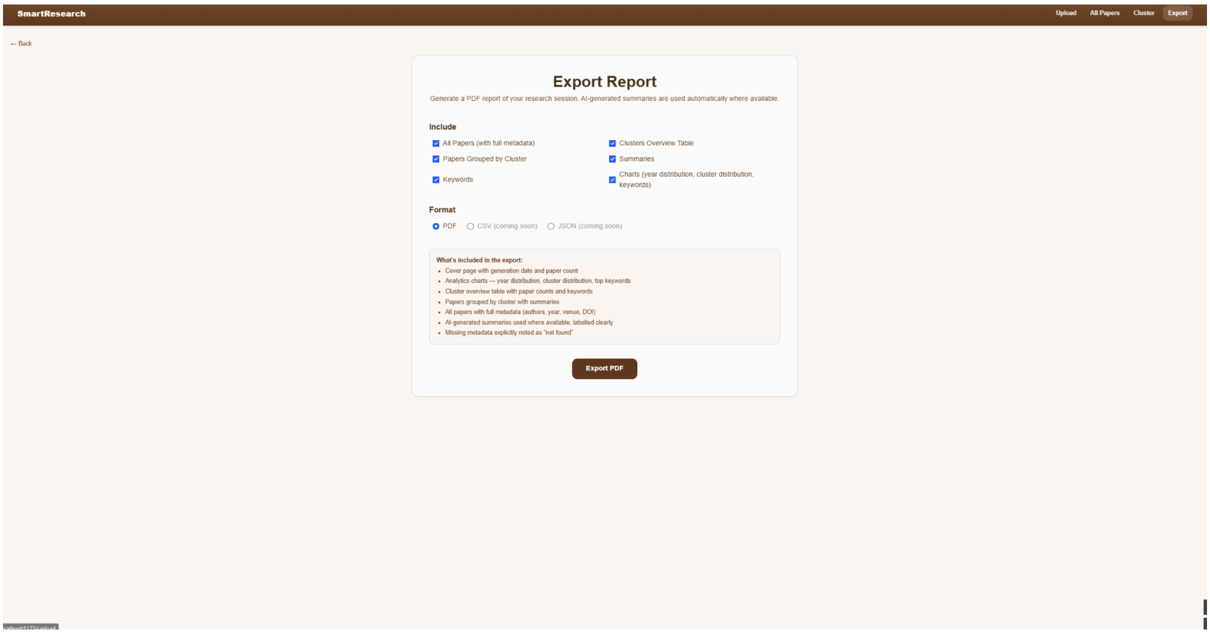

Figure 18: Export Page

10.3 Export Content Options

Option	Description
All Papers	Includes the full processed paper list.
Clusters List	Includes generated cluster overview.
Papers by Cluster	Includes papers grouped under clusters.
Charts	Includes cluster distribution and keyword charts
Recommendations	Includes Semantic Scholar recommended papers where available
Summaries	Includes summary information where available.
Keywords	Includes generated keywords where available.
Cluster Insights	Includes cluster-related information where available.

10.4 Export Format Options

Format	Interface Option	Final Support Notes
PDF	Visible in interface.	Primary supported export format.
CSV	Visible in interface.	Not implemented. Backend returns error if selected.
JSON	Visible in interface.	ot implemented. Backend returns error if selected.

PDF was the confirmed supported export format during final verification. CSV and JSON may appear in the interface, but users should rely on PDF unless additional backend testing confirms support for those formats.

10.5 Generate a PDF Export

Step	Action	Expected Result
1	Open Export page.	Export interface appears.
2	Keep or adjust selected content options.	Selected options remain checked.
3	Select PDF format.	PDF radio button is selected.
4	Click Export.	Export request is sent.
5	Wait for download.	PDF file downloads if successful.
6	Open downloaded PDF.	Export report displays selected content.

Papers by Cluster

Philosophy in Dialogue with Other Sciences on the Example of Albert the Great and Thomas Aquinas

Keywords: philosophy, interdisciplinarity, cybernetics, schmidt, sciences, artificial, science, intelligence

5 paper(s) in this cluster

• Philosophy in Dialogue with Other Sciences on the Example of Albert the Great and Thomas Aquinas

Separation of natural sciences from philosophy and the development of empirical sciences were long processes that led to mistrust and even opposition between philosophers and naturalists. This paper responds to the question how to re-establish dialogue between philosophy and other sciences. Authors propose a solution in the so-called model of the dialogically open philosophy of two famous philosophers and theologians Albert the Great and Thomas Aquinas.

• Interdisciplinary Description of Complex System

Philosophers were reluctant to address interdisciplinarity during the 20" century. But things have changed in the 21" century, since a two-level relationship between philosophy and interdisciplinarity has been established: philosophy of interdisciplinarity and philosophy as interdisciplinarity. Thus far scholars have shown more interest in exploring the first level of that relationship. The aim of this article is to closely examine the developmental path of a philosophy of interdisciplinarity envisioned and constructed by Jan Cornelius Schmidt in the past two decades. In our opinion, it has reached two milestones.

• The Philosophy of Cybernetics

This special edition of The Journal on Systemics, Cybernetics, and Informatics (JSCI), "Philosophy & Cybernetics", would not be complete without describing the link between the two words of the subtitle. "Interdisciplinary" is the heart of this journal and its embracing organization, the International Institute of Informatics and Systemics (IIIS), one of the few, if not the only organization, devoted to the study of the concept. We find "cybernetics" within the context of "interdisciplinary", also carrying the same import of integration, interdependence, and organicity as "cybernetics". To see why requires thinking about thinking, or philosophy, more precisely a philosophical system. Yet, all philosophical systems are beset by the insurmountable problem of attaining universal incontrovertible truths, that is, metaphysical certainty.

• Philosophy, the Humanities & the Life of Freedom

Abstract Humanistic disciplines have family resemblances rather than a simple shared common aim or method, and, like literal family resemblances, these have an explanation that comes from their historical relationships to one another. Philosophy, in particular, is closely connected to the sciences it has spun off over the centuries, but remains distinct from them, because normative inquiry uses methods different from those of any contemporary science. But much philosophical inquiry, like much humanistic work, is also idiographic rather than nomothetic; it focuses our attention on particular things, rather than seeking generalizations.

• Research Paper On Artificial Intelligence And It's Applications

In the future, intelligent machines will replace or enhance human capabilities in many areas. Artificial intelligence is the intelligence exhibited by, machines or software. It is the subfield of computer science. Artificial] intelligence is becoming a popular field in computer science as it has| enhanced the human life in many areas. Artificial intelligence in the last two decades has greatly improved performance of the manufacturing, service] sector and so in the field of education.

BERT: Pre-training of Deep Bidirectional Transformers for

Figure 19: Downloaded Export Report

10.6 Export Error Message

Possible Cause	Action

Backend not running.	Restart backend and try again.
No papers uploaded.	Upload papers first.
Unsupported format selected.	Use PDF unless CSV/JSON support has been confirmed in the final build.
Backend export error.	Check backend terminal for error messages.
Export payload mismatch.	Confirm frontend/backend export payload compatibility.
Browser blocked download.	Check browser download permissions and retry.

---

11. Deleting Papers

11.1 Purpose of Delete

The delete function removes a stored paper from the SmartResearch local corpus. This is useful for clearing test documents, removing incorrect uploads, or resetting the paper collection.

11.2 Delete a Paper

Step	Action	Expected Result
1	Open All Papers.	Paper table appears.
2	Find the paper to remove.	Target paper row is visible.
3	Click Delete.	Browser confirmation prompt appears where implemented.
4	Confirm deletion.	Paper is removed from the system.
5	Wait for refresh.	Updated paper list appears.

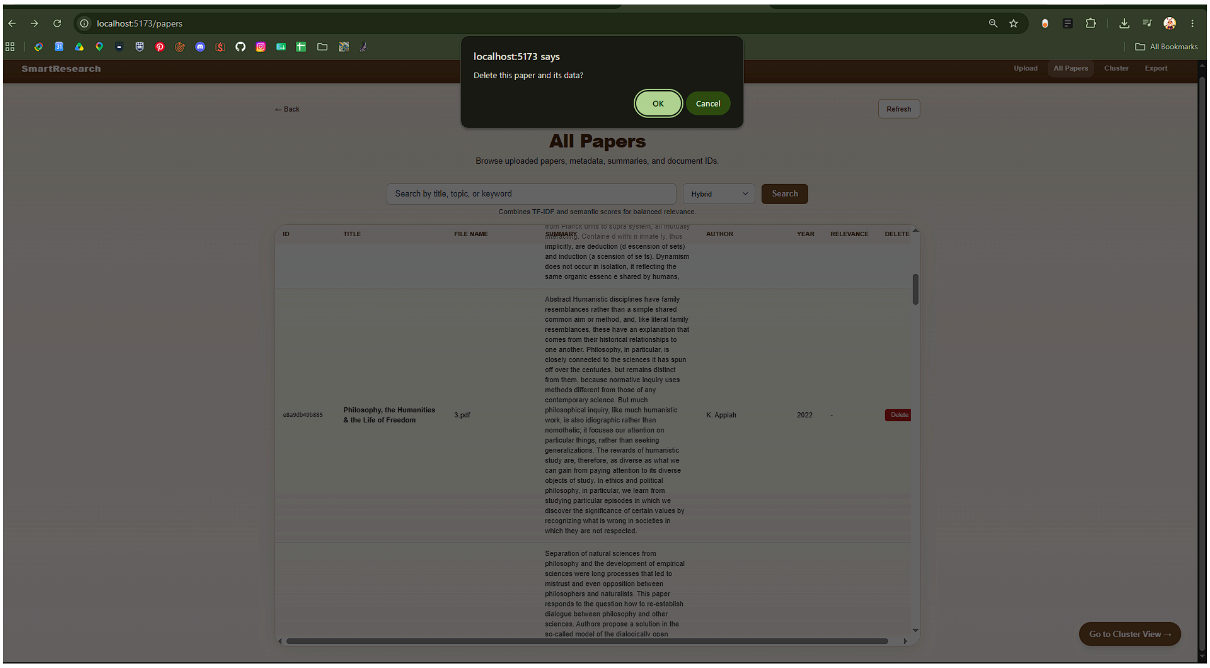

Figure 20: Delete Confirmation Prompt

11.3 Delete Warning

Deleting a paper removes the stored paper and associated local data where available. Users should only delete papers they no longer need in the current local prototype.

12. Recommended End-to-End Workflows

12.1 Standard User Workflow

Stage	User Action	SmartResearch Feature
Collect Papers	Gather academic PDFs.	External to system.
Add Papers	Upload PDFs.	Upload page.
Process Papers	Wait for upload and processing to complete.	Backend ingestion pipeline.
Review Papers	Browse uploaded papers.	All Papers page.
Inspect Details	Open individual paper records.	Detail modal.
Retrieve Papers	Search using keyword, semantic, or hybrid mode.	All Papers search.
Explore Themes	Review generated clusters.	Cluster page.
Generate Summary	Generate an abstractive summary	Full Summary page
Recover/Replace Summary	Overwrite existing summary, or recover extracted summary	Full Summary page.
Export Output	Generate report.	Export page.
Verify Manually	Check source papers before academic use.	User judgement.

12.2 Suggested First-Time Workflow

For a first test run:

Start backend.
Start frontend.
Upload two or three academic PDFs.
Confirm upload status reaches completed.
Open All Papers.
Open one paper detail modal.
Run one keyword search.
Run one semantic search.
Run one hybrid search.
Open Cluster page.

Open one cluster modal.
Export a PDF report.
Open downloaded report.
Delete one test paper if cleanup is needed.

12.3 Larger Paper Set Workflow

Step	Action	Reason
1	Start with a smaller upload first.	Confirms the system is running properly.
2	Upload the larger batch.	Adds complete paper set.
3	Wait for processing.	OCR and embeddings may take time.
4	Refresh All Papers.	Confirms stored corpus.
5	Run representative searches.	Checks retrieval behaviour.
6	Review clusters.	Checks thematic grouping.
7	Export results.	Produces reusable output.

---

13. Troubleshooting

13.1 Common Issues

Problem	Likely Cause	Recommended Action
Frontend does not load.	Frontend server is not running.	Run npm run dev in the frontend folder.
Upload fails.	Backend not running, invalid file type, or backend error.	Restart backend and confirm the file is a PDF.
Upload stays processing.	Large PDF, OCR fallback, or backend processing delay.	Wait, then check backend terminal output.
No papers appear in All Papers.	No successful uploads or backend storage is empty.	Upload a PDF and click Refresh.
Metadata is missing or incomplete.	PDF metadata is not extractable or enrichment is unavailable.	Use filename/summary fallback and check the original paper.
Summary is missing.	Text extraction or summary generation failed.	Try another PDF and check backend logs.
Search fails.	Backend unavailable or search endpoint error.	Restart backend and retry.
Cluster page is empty.	No papers, missing semantic data, or clustering error.	Upload papers and refresh or reindex if available.
Similar papers or recommendations are unavailable.	Not enough uploaded papers, missing metadata, or external service unavailable.	Upload more papers, check DOI/title metadata, or treat as unavailable.
Export fails.	Backend export error, no papers uploaded, or unsupported format selected.	Use PDF and check backend logs.
Delete fails.	Backend delete endpoint error.	Refresh and retry.

13.2 Backend Not Running

Symptoms:

Symptom	Meaning
Upload fails immediately.	Frontend cannot reach backend.
Search does not work.	API unavailable.
Cluster page does not load clusters.	Backend endpoint unavailable.
Export fails.	Export endpoint unavailable.

Fix:

```
cd backend
source .venv/bin/activate
uvicorn app:app --reload
```

Then refresh the frontend.

13.3 Frontend Not Running

Symptoms:

Symptom	Meaning
Browser cannot open the SmartResearch interface.	Vite frontend server is not running.
Terminal does not show local frontend URL.	npm run dev has not started or failed.

Fix:

```
cd frontend
npm run dev
```

13.4 Upload Takes Too Long

Cause	Explanation
Large PDF	More content takes longer to extract and process.
Scanned PDF	OCR fallback may be slower.
Semantic Indexing	Embedding generation may take time.
First Run	Model/dependency loading can be slower initially.

Recommended action: Wait for the backend to finish processing, then check All Papers. For testing, use smaller text-selectable PDFs first.

13.5 Search Returns No Results

Possible Cause	Action
Query too narrow.	Try broader terms.
Wrong search mode.	Try Hybrid or Semantic.
No matching papers.	Upload more relevant papers.
Backend search issue.	Check backend terminal for errors.
Stored text missing.	Confirm text extraction worked.

13.6 Clusters Do Not Appear

Possible Cause	Action
No papers uploaded.	Upload papers first.
Papers not fully processed.	Wait and refresh.
Semantic index missing.	Use reindex endpoint if available.
Backend cluster error.	Check backend terminal.
Too few documents.	Upload more papers for more meaningful clustering.

13.7 Export Does Not Download

Possible Cause	Action
Backend not running.	Restart backend.
No processed papers.	Upload papers first.
Unsupported format selected.	Use PDF unless CSV/JSON support is confirmed.
Backend export crash.	Check terminal logs.
Browser blocked download.	Check browser download permissions.

13.8 OCR Does Not Work

Possible Cause	Action
OCR dependencies not installed.	Check Tesseract/pdf2image setup.
PDF scan quality is poor.	Try a clearer PDF.
PDF is image-heavy.	Expect slower or weaker extraction.
Backend error occurred.	Check terminal logs.

---

14. Known Limitations for Users

Limitation	User Impact
PDF-only upload	Users cannot upload Word, PowerPoint, image, or spreadsheet files as papers.
Scanned PDFs may vary	OCR fallback depends on scan quality and local OCR setup.
Metadata may be incomplete	Some papers may not show full title, author, year, DOI, venue, abstract, or external metadata.
Metadata confidence depends on available information	Reliability indicators depend on extractable PDF data, DOI/title quality, and external metadata availability.
Summary output may vary	Summaries may not always follow a strict academic structure.
Abstractive summary may be slow	DistilBART summarization depends on document length and CPU or GPU utilization.
Search scores are guidance only	Relevance scores should not replace manual review.
Cluster labels may be rough	Generated cluster keywords may need human interpretation.
Similar-paper results depend on the uploaded corpus	Related-paper results work best when multiple relevant papers have been uploaded.
External recommendations may vary	External recommendations depend on metadata quality, network access, and third-party service availability.
Export format support depends on final backend implementation	PDF is the primary export pathway unless CSV/JSON support is confirmed by final testing.
No user accounts	The prototype does not support login or user-specific libraries.
Local-only system	The system runs locally and is not deployed as a hosted web application.
No full production security layer	Users should avoid uploading sensitive or restricted documents without considering local storage implications.
Disabled or unfinished controls may appear	Some interface controls may indicate future functionality rather than final working features.

15. User Safety and Academic Good Practice

15.1 Data Handling

SmartResearch stores uploaded files and generated outputs locally. Users should only upload documents they are permitted to store and process on the machine running the prototype.

15.2 Academic Use

SmartResearch outputs should be used as research support, not as final academic evidence on their own.

Users should treat SmartResearch outputs as support material for screening and organisation, not as final academic evidence. Important claims, summaries, metadata, search results, recommendations, and cluster groupings should be checked against the original papers before being cited or relied on.

Output	Correct Use
Summary	First-pass screening and review support.
Metadata	Paper identification support, subject to checking.
Search Results	Retrieval aid for finding relevant papers.
Clusters	Thematic exploration guide.
Export Report	Reusable note/report output.

15.3 Recommended Use

Use SmartResearch to:

Good Use	Reason
Screen paper collections faster.	Summaries help with first-pass review.
Find papers by topic or keyword.	Search improves retrieval.
Explore themes across a corpus.	Clustering supports topic grouping.
Prepare review material.	Export creates reusable output.

Do not use SmartResearch to:

Avoid	Reason
Replace reading original papers entirely.	Generated summaries may miss nuance.
Cite claims without checking sources.	Outputs may be incomplete or imperfect.
Treat cluster labels as final academic categories.	Clusters are generated automatically.
Upload sensitive files without checking storage expectations.	Prototype uses local storage without account/security controls.

Appendix A: Quick Reference

A.1 Main Pages

Page	Route	Purpose
Upload	/upload	Upload PDFs.
All Papers	/papers	Browse, search, inspect, and delete papers.
Cluster	/cluster	View generated topic clusters.
Export	/export	Generate export output.

A.2 Main User Actions

Action	Where
Upload PDF	Upload page.
Preview PDF	Upload page after upload completion, where available.
View Papers	All Papers page.
Search Papers	All Papers page.
View Metadata/Summary	Paper detail modal.
Generate AI summary	Full summary page
Recover/Replace Summary	Full summary page
View Clusters	Cluster page.
View Cluster Papers	Cluster modal.
Export Results	Export page.
Delete Paper	All Papers page.

A.3 Recommended Search Mode

Situation	Recommended Mode
Exact word, title, or phrase known.	Keyword
Concept known but wording unknown.	Semantic
General search across uploaded papers.	Hybrid

A.4 Common Buttons

Button	Purpose
Go to All Papers	Opens paper review table.
Refresh	Reloads stored papers.
Search	Runs selected search mode.
Show All Papers	Clears search view and reloads all papers.
Go to Cluster View	Opens Cluster page.
View Papers	Opens cluster detail modal.
Generate Summary	Generates an abstractive summary for the document
Short/Medium/Long	Selects target length for abstractive summary
Replace Summary	Replaces current summary with generated abstractive summary
Recover Summary	Recovers and replaces current summary with extracted summary
Next	Opens Export page where available.
Export	Generates export output.
Delete	Removes a stored paper.
Delete All	Removes all stored papers at once

Appendix B: Example End-to-End Walkthrough

Scenario: Reviewing a Small Paper Set

A student has collected five academic papers for a literature review and wants to organise them before writing.

Step	Action	Result
1	The student starts the backend and frontend.	SmartResearch opens in the browser.
2	The student uploads five PDF files.	Upload progress appears.
3	The student waits for processing to complete.	The files show a completed status.
4	The student opens **All Papers**.	The uploaded papers appear in the table.
5	The student clicks a paper title.	The paper detail modal opens, showing the available metadata and extractive summary.
6	The student clicks **Generate Summary**.	The Full Summary page opens.
7	The student selects **Short**, **Medium**, or **Long**, then clicks **Generate Summary**.	An optional abstractive summary is generated. Processing may take longer when a GPU is not available.
8	The student reviews the generated summary and clicks **Replace Existing Summary** where required.	The generated summary replaces the existing summary in the All Papers view.
9	The student returns to **All Papers** and searches for a topic.	Relevant papers appear in the search results.
10	The student switches between keyword, semantic, and hybrid search modes.	Results can be compared using different retrieval methods.
11	The student opens the **Cluster** page.	Papers are grouped into generated topic clusters.
12	The student reviews the cluster keywords and paper groupings.	Topic themes and relationships between papers can be identified.
13	The student opens the **Export** page.	Export options are displayed.
14	The student selects the required report content and exports a PDF report.	The report downloads for external review.

15	The student manually checks the original papers before using the exported material in written work.	SmartResearch is used as a research support tool rather than a replacement for academic judgement.

Where required, the student can return to the paper detail modal and click Recover Summary to restore the original extractive summary.